The biological and chemical processes that take us from a cell to a baby really are **absolutely incredible**. As soon as your pregnancy begins, your **amazing body** transforms so it can protect this **new life**, while the new life itself becomes a fully functioning human being, **one cell at a time** but at remarkable speed.

From cell to baby

The zygote of life

Your baby begins life as a single cell, created by a sperm fusing with an egg. This organism contains all the potential for new life—a full set of chromosomes (23 pairs, one in each pair from each parent). These carry your baby's unique blueprint encoded in genes. Although we often refer colloquially to this earliest stage of life as an embryo, in fact we're a few steps from an embryo yet. For the time being, your baby is a zygote. The zygote travels down the fallopian tube and as it travels, it is already subdividing and multiplying to make 16 cells. It's surrounded by a protective membrane called the zona pellucida, which has a very special function—it prevents the zygote from implanting in the fallopian tube, or anywhere that isn't your uterus.

Morula to blastocyst

During its journey down the fallopian tube to the uterus, the zygote becomes known as a morula (because the cells are grouped together and look like a mulberry, from the Latin morus). The morula is a hive of activity: its cells subdivide rapidly and eventually create a kind of bubble structure with an outer membrane (the trophoblast) containing a mass of cells (the endoderm) within a cavity. Amazingly, that bundle of cells sitting against the membrane wall will become the key to your baby's breathing and digestion, among other things. At this stage your baby becomes known as a blastocyst. The blastocyst floats around in the uterus for a while, then sends chemical messages that break down the zona pellucida. Now exposed, the "sticky" trophoblast latches onto and implants itself in the uterine wall. All this happens around seven to nine days after fertilization. Your baby has now doubled in size since fertilization and is made up of more than a hundred cells. Once implanted, the blastocyst is enveloped by the uterine lining (endometrium) and snuggles down deep within the cells of the lining. It can now receive nourishment directly from your uterus.

Embryo to fetus

At four weeks, the blastocyst can be called an embryo. The trophoblast has started to develop into the placenta (see p.154), and secretions from your uterus are nourishing the cells so that they divide and subdivide and begin to form your baby's organs, skeleton, and features. Crucially, the embryo is made up of three layers; each will develop into specific parts of your baby's body. Now that your baby is being nourished by your body, the business of becoming more "babylike" can begin. At about 10 weeks, when his facial features are in place and most of the major organs are on their way to being formed, your baby has become a fetus—the medical name he keeps until birth.